LAB 3

#1

```
#include <iostream>
using namespace std;

class Demo
{
    int x;
public:
    Demo(int x)
    {
        this->x = x;
    }
    void show()
    {
        cout<<this->x<<"\n";
    }
};

int main()
{
    Demo obj(10);
    Demo *p = &obj;

    obj.show();
    p->show();
}
```

```
PS C:\Users\lenovo\Desktop\DSA>
10
10
```

#2

```
#include <iostream>
using namespace std;

class B;

class A
{
    int x;
```

```
PS C:\Users\lenovo\Desktop\DSA>
20 10
```

```
public:
        A(int x){ this->x=x; }
    friend void swap(A&,B&);
    void show(){ cout<<x<<" "; }
};

class B
{
    int y;
public:
        B(int y){ this->y=y; }
    friend void swap(A&,B&);
    void show(){ cout<<y<<" "; }
};

void swap(A &a,B &b)
{
        int
    t=a.x;
    a.x=b.y;
    b.y=t;
}

int main()
{
        A a(10);
        B b(20);
        swap(a,b);
        a.show();
        b.show();
}
```

#3

```
#include <iostream>
using namespace std;

class B;

class A
```

```
PS C:\Users
30
```

```
{
    int x;
public:
        A(int x){ this->x=x; }
    friend int add(A,B);
};

class B
{
    int y;
public:
        B(int y){ this->y=y; }
    friend int add(A,B);
};

int add(A a,B b)
{
        return a.x+b.y;
}

int main()
{
        A a(10);
        B b(20);
        cout<<add(a,b);
}
```

#4

```
#include <iostream>
using namespace std;

class A
{
    int x;
    friend class B;
public:
    A(int x){ this->x=x; }
};
```

```
PS C:\Users\lenovo\Desktop\DSA>
50
```

```
class B
{
public:
        void show(A a)
        {
                cout<<a.x;
        }
};

int main()
{
        A a(50);
        B b;
    b.show(a);
}
```

#5

```
#include <iostream>
using namespace std;

class Rect
{
    int l,b;
public:
    void set(int l,int b)
    {
        this->l=l
        ;
        this->b=b
        ;
    }
    int area()
    {
        return l*b;
    }
};

int main()
{
```

```
PS C:\Users\lenovo\Desktop\DSA>
 ; if ($?) { .\tempCodeRunnerFil
6 20 42
```

```
        Rect r[3];
    r[0].set(2,3);
        r[1].set(4,5);
        r[2].set(6,7);

            for(int i=0;i<3;i++)
        cout<<r[i].area()<<" ";
}
```

#6

```
#include <iostream>
using namespace std;

inline int cube(int x)
{
    return x*x*x;
}

int main()
{
    int n;
    cin>>n;
    cout<<cube(n);
}
```

```
PS C:\Users\lenovo\Desktop\DSA>
 ; if ($?) { .\tempCodeRunnerFil
4
64
```

#7

```
#include <iostream>
using namespace std;

class A
{
public:
    int x;
    A(int x=0){ this->x=x; }
};

A modify(A a)
{
```

```
PS C:\Users\len
15
```

```
        a.x+=10;
        return a;
}

int main()
{
        A a(5);
        A
    b=modify(a);
    cout<<b.x;
}
```